

Deep Learning Quiz 2

Total points 4/5

The respondent's email (**f20230458@goa.bits-pilani.ac.in**) was recorded on submission of this form.

- ✓ Early perceptron learning updates weights only when a prediction is wrong (a binary error signal). *1/1

Modern neural networks instead use **loss functions**, which produce continuous error values.

Why is this shift important?

(Select all that apply)

- ☒ A) Continuous loss values change smoothly as predictions improve, while binary error changes abruptly ✓
- ☒ B) Binary right/wrong error does not say how wrong a prediction is ✓
- ☐ C) Loss functions guarantee finding the global minimum
- ☒ D) Continuous loss allows the model to keep learning even when its predictions are close to correct ✓

Feedback

- A ✓ Correct — Loss gradients indicate how to change weights.
- B ✓ Correct — Binary error lacks magnitude information.
- D ✓ Correct — Small errors still produce useful learning signals.



✗ Consider these two training rules: *0/1

Perceptron learning: updates weights only when a mistake occurs

Gradient-based learning: updates weights based on the gradient of a loss function

Which statements are correct?

(Select all that apply)

- ☒ A) Perceptron learning uses only the sign of the error ✓
- ☒ B) Gradient-based learning uses information about the size of the error ✓
- ☒ C) Perceptron learning can be fully online (can update weights one example at a time, without storing the full dataset) ✓
- ☒ D) Gradient-based learning can be applied to multi-layer networks ✓
- ☐ E) Gradient-based learning has higher memory requirement because it requires storing the full dataset in memory

Correct answer

- ☒ A) Perceptron learning uses only the sign of the error
- ☒ B) Gradient-based learning uses information about the size of the error
- ☒ C) Perceptron learning can be fully online (can update weights one example at a time, without storing the full dataset)
- ☒ D) Gradient-based learning can be applied to multi-layer networks
- ☒ E) Gradient-based learning has higher memory requirement because it requires storing the full dataset in memory



Feedback

C  Incorrect – Perceptron learning only needs the current example to make an update, so it can work in a streaming setting.

✓ Linear regression with squared loss has a closed-form (analytical) solution. *1/1

Multi-layer neural networks are trained using iterative optimization instead.

Which reason best explains this difference?

(Single correct)

- ☐ A) Neural networks have too many parameters and cannot be written down with formulas
- ☒ B) Nonlinear activations make the loss function difficult or impossible to solve analytically ✓
- ☐ C) Iterative methods are always faster than analytical solutions
- ☐ D) Analytical solutions require storing all gradients in memory

Feedback

B  Correct – Nonlinearities break the assumptions needed for analytical solutions.



✓ **Why backpropagation is necessary.** *

1/1

- ☒ A) Without an activation function, stacking multiple layers of neurons does not increase the model's expressive power ✓
- ☒ B) Activation functions allow neural networks to represent non-linear input-output relationships ✓
- ☐ C) Depth alone (more layers) makes a network nonlinear, even without activation functions
- ☒ D) A network without activation functions is equivalent to a single linear model ✓

Feedback

A ✓ *Correct — Without activation functions, the entire network is just a composition of linear operations.*

B ✓ *Correct — Activation functions introduce nonlinearity, which is essential for modeling complex relationships.*

D ✓ *Correct — Multiple linear layers collapse into one equivalent linear transformation.*



- ✓ The Pitts neuron and the Perceptron both compute a weighted sum of inputs followed by a threshold decision. *1/1

The key difference between them is how the weights are obtained.

Which statement best captures this difference?(Single correct)

- ☐ A) Pitts neurons use binary weights, while perceptrons use continuous weights
- ☒ B) Pitts neurons have fixed weights, while perceptrons learn weights from data ✓
- ☐ C) Perceptrons can model nonlinear functions, while Pitts neurons cannot
- ☐ D) Pitts neurons are trained using Hebb's training rule, while perceptrons are trained using perceptron training rule.

Feedback

B ✓ Correct — Pitts neurons are hand-designed; perceptrons learn weights from data.

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